

- (ii) reflex agents with state :-
- tracks environment to do reflex actions
- (iii) Goal-based agent :-
- chooses actions to achieve a specific goal
 - Uses planning & search.
- (iv) Utility-Based Agent :-
- selects agent actions that maximize utility
 - Considers trade-offs & performance.

28/01/26, Wednesday

*. Problems, Problem space and search :

⇒ To make a model of AI :

- ① define the problem precisely
- ② analyse the problem
- ③ isolate and represent the task knowledge.
- ④ choose the best problem solving technique.

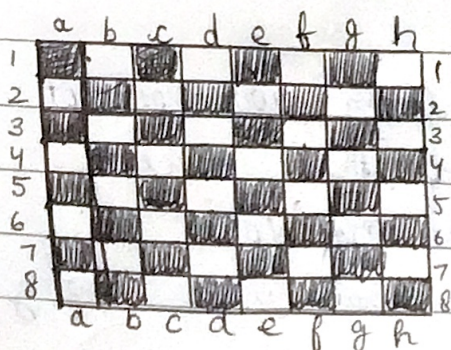
⇒ Define the problem as a state space search.

- ① initial state
- ② goal state
- ③ legal state

⇒ examples :

- ① Chess problem : It has $(10)^{20}$ rules.

Rules :



white (pauen)

square (file e, rank 2)

square (file e, rank 3)

square (file e, rank 4)

is empty

sq. (file e, rank 2)

sq. (file e, rank 4)

② Water jug problem :



- given :
- 2 mugs
 - 4 gallon , 3 gallon
 - no measuring marker
 - a pump to fill jugs

goal : 2 gallon of water in 4 gallon jug.

$\Rightarrow$ state space : ordered pair $(x, y) \Rightarrow x = 1, 2, 3, 4$
 $y = 1, 2, 3$

$\Rightarrow$ start state : $(0, 0)$

$\Rightarrow$ goal state : $(2, n)$

NOTE : Production rules are not the sequence of problem solving its just rules.

$\Rightarrow$ Production rules for water jug problem :

① $(x, y) \rightarrow (4, y)$ fill the 4 gallon jug.
 if $x < 4$

② $(x, y) \rightarrow (x, 3)$ fill the 3 gallon jug.
 if $y < 3$

③ $(x, y) \rightarrow (x-d, y)$ pour some water out of 4 gallon jug.
 if $x > 0$

④ $(x, y) \rightarrow (x, y-d)$ pour some water out of 3 gallon jug.
 if $y > 0$

⑤ $(x, y) \rightarrow (0, y)$ empty the 4 gallon jug.
 if $x > 0$

⑥ $(x, y) \rightarrow (x, 0)$ empty the 3 gallon jug.
 if $y > 0$

⑦ $(x, y) \rightarrow (4, y-(4-x))$ fill 4 gallon & fill 3 gallon to the difference in x .
 if $(x+y \geq 4 \text{ \& } y > 0)$

⑧ $(x, y) \rightarrow (x-(3-y), 3)$
 if $(x+y \geq 3 \text{ \& } x > 0)$

$$(9) \quad (x, y) \longrightarrow (x+y, 0) \\ \text{if } (x+y \leq 4 \text{ \& } y > 0)$$

$$(10) \quad (x, y) \longrightarrow (0, x+y) \\ \text{if } (x+y \leq 3 \text{ \& } x > 0)$$

$$(11) \quad (x, y) \longrightarrow (2, 0) \\ \text{if } (0, 2)$$

$$(12) \quad (x, y) \longrightarrow (0, y) \\ \text{if } (2, y)$$

4 gallon 3 gallon Rule applied.

0

0

2

0

3

9

3

0

2

3

3

7

4

2

5

0

2

11

2

0

goal achieved.

